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FIRE SAFE - GREEN & COMPACT TRANSFORMER

Advanced Dielectric Fluids

at

SUPER SHAKTI

By

SAIKAT KOLEY

SAVITA OIL TECHNOLOGIES LIMITED

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Savita Oil Technologies Limited

Presentation on:

Use of Natural Ester Fluids in Transformers and allied accessories

GREEN PERFORMER

SAVSOL | TRANSOL

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efficient | safe | eco-friendly

Transformer Oil

SAVITA has successfully supplied Transformer Oil to the complete class range of Distribution & Power Transformers.

For 765 KV Power Transformers, we have completed projects with following OEM's :

- ABB
- GE T & D
- CG POWER
- SIEMENS
- BHEL
- TBEA
- T&R

Approvals

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SAVITA's TRANSOL has been supplied directly to Transformer Manufacturer's in over 75 countries.

Besides complying with International Standards like IEC, BS, ASTM TRANSOL has been approved by most local major utilities, some examples :

- India: Power Grid Corporation of India Limited(PGCIL)
- Saudi Arabia : Saudi Electric Company (SEC)
- UAE : DEWA, SEWA , ADWEA (Dubai, Sharjah and Abu Dhabi Authorities)
- Malaysia: Tanega Nasional Berhard (TNB)
- Turkey: Turkey Electricity Transmission Company (TEIAS)
- Indonesia: Perusahaan Listrik Negara (PLN)
- Nigeria: Electricity and Power Authority
- Oman: Oman Electricity and Transmission Company (OETC)
- Bangladesh: Bangladesh Power Development Board
- Sri Lanka: Ceylon Electricity Board

Testing Facilities

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- We have In-house testing laboratories at all our manufacturing plants, which are accredited by NABL (National Accreditation Board for Testing and Calibration Laboratories)

- Our Products are tested in International Laboratories for Electrical Power and Energy Technology such as Laborelec-Belgium, KEMA-Netherland, Dobles-USA & Terna-Italy.

Green footprint

With an Installed capacity of ***54.15 MW,*** Savita generates 110 million units of power in a year.

Savita's renewables capacity displaces approximately 80,000 tons of carbon dioxide, making it one of the **only carbon positive oil manufacturing companies** in Asia.

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Natural Ester Fluid

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Let’s Begin with a Few Questions...

🔍 1. How often do you change or top up mineral oil in your transformers?

2. Are you facing frequent BDV drops due to moisture issues?

3. Do you see performance challenges in high-temperature or humid areas?

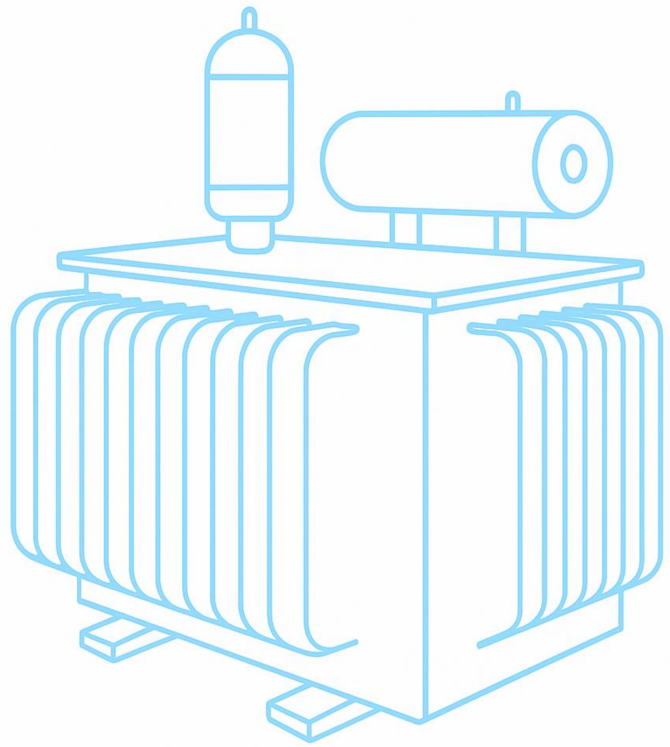
4. Are you satisfied with the current life span of your transformers?

🔥 5. Have there been any safety concerns related to fire risk or oil leakage?

6. Is oil disposal becoming a compliance or cost burden for your team?

📋 7. Do maintenance audits highlight issues linked to oil performance?

♻️ 8. Have you explored retrofitting to improve performance and sustainability?



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Brief History of Transformer Oils

- In the late 1880s Veg oils were used as dielectric fluids
- Soon found to be useless for free breathing transformers
- Mineral Oils made appearance next
- This was followed by Polychlorinated Biphenyls (PCBs)
- Silicon came into use by 1970s as an alternative to PCBs
- Since 1990s, Veg oils came back due to its GREEN credentials
- Synthetic Esters are also in vogue today

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Key Factors for Transformer Value Chain

Life

- It is important that the Transformer is up and running at all times.
- The shelf Life of the Transformer should be **>20 Yrs for DT & >25 Yrs for PT**

Maintenance

- Minimum **preventive and breakdown Maintenance**

Safe

- **Safe for operation** in all aspect and does not create any Safety hazard while in operation.
- **Fire Safe**

Footprint

- **Small Footprint** meeting all relevant standards.

Environmental

- **Reduced environmental impact** as much as possible.
- **Carbon Footprint** should be very less (**Carbon Neutral**)

With our 100+ years of experience Mineral Oil filled Transformer is not the proper answer for above value chain Factors.

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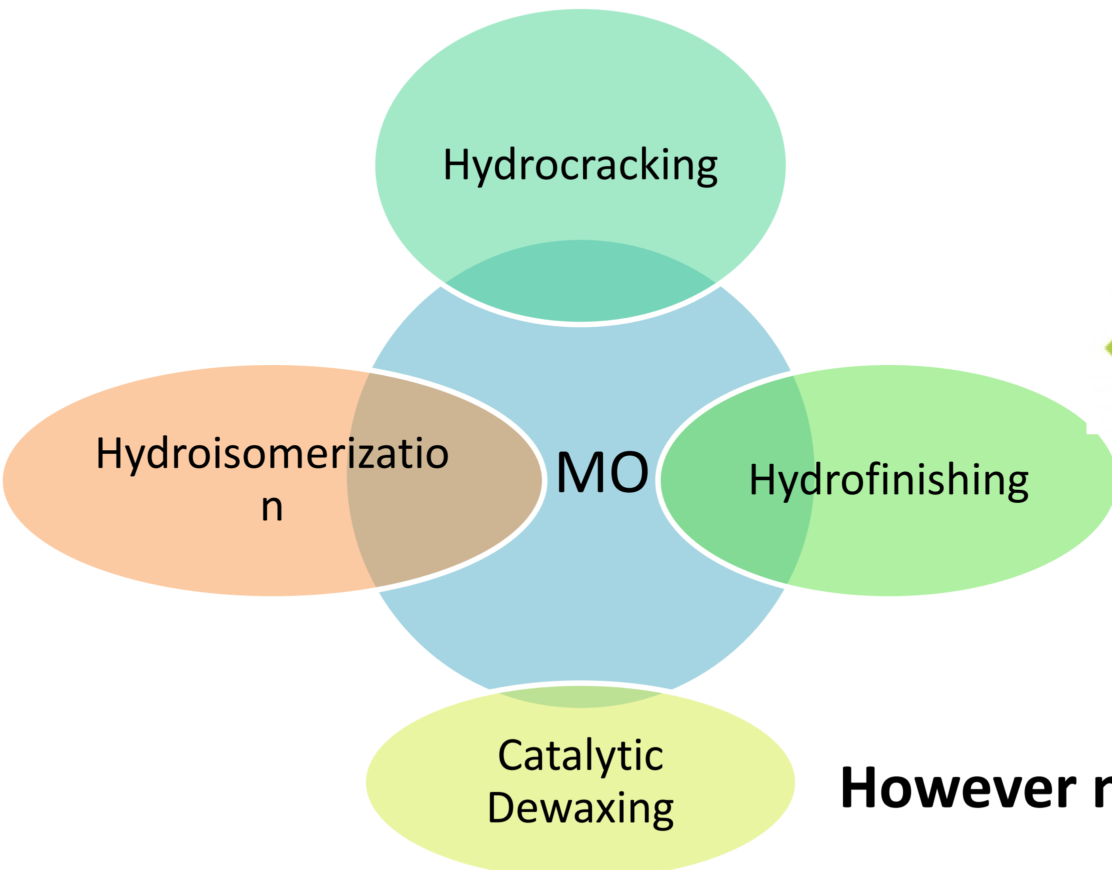
Advancement of MO manufacturing Technologies



- **Oxidation Stability**
- **Thermal Stability**
- **Low Temperature properties**
- **Low viscosity**

However no Change in Flash Point & Fire Point

Base Material for Natural Ester

Major Vegetable Seed Oils

- Safflower, Sunflower
- Mustard, Canola
- Soybean
- Linseed ,Tung Oil
- Cotton Seed Oil
- Tobacco Seed Oil

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FAILURE AND TROUBLE EXPERIENCE - TRANSFORMER

Cause	Impacted System	Percent of Reported Failures
Insulation failure	Dielectric	26%
Manufacturing failure	Unknown	24%
Unknown	Unknown	16%
Loose connections	Current Carrying	7%
Improper maintenance	Unknown	5%
Overloading	Dielectric	5%
Oil contamination	Dielectric	4%
Line surges	Dielectric/Mechanical	4%
Fire/explosions	Dielectric/Containment	3%
Lightning	Dielectric	3%
Floods	Containment	2%
Moisture	Containment	1%
Total		1.00%

~40%

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Major Parameters

	Parameter	Mineral Oil (IS 335:2018)	Natural Esters
1	Flash point °C	135 (min)	265
2	Pour point °C	-10 /-40	-18 to -21
3	Density@20°C gm/cm^3	0.895 Max	0.92
4	Viscosity mm^2/sec 100°C	-	7.2-8.5
5	Viscosity mm^2/sec 40°C	15/12 Max	32-35
6	BDV kV/2.5mm gap	70	70
7	Tan delta @90°C	0.005	0.02-0.03
8	Acidity mg KOH/gm	<0.01	0.03
9	Water content mg/kg	<40	>50
10	Biodegradation	Not readily biodegradable	Yes

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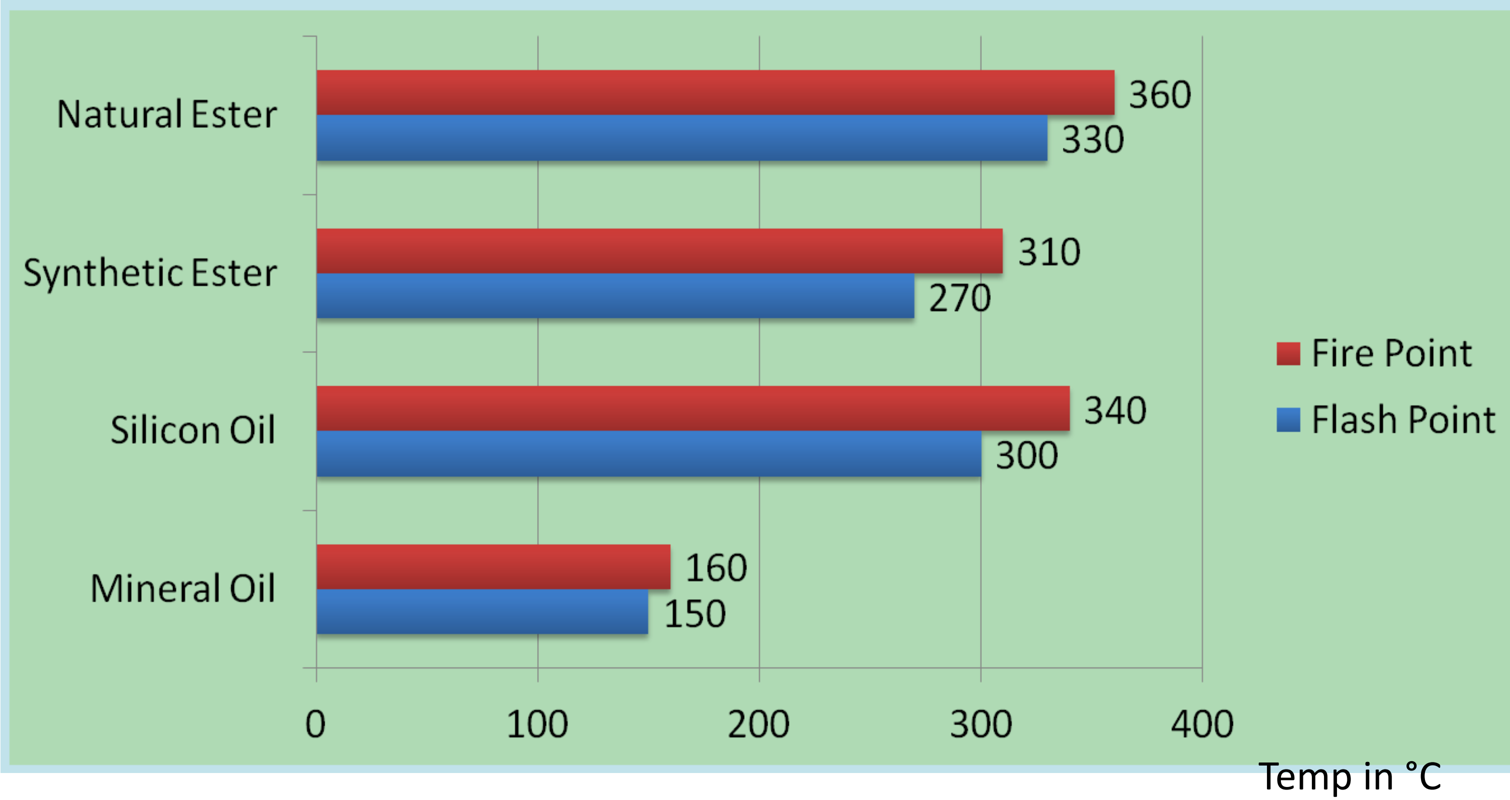
bioTRANSOL for Achieving better fire safety

High Flash(>250°C) & Fire Point(>300°C)
'K' class Fluid as per IS 13503:2013



Class	Fire Point
O	≤ 300°C
K	> 300°C
L	No Measurable Fire Point

Class	Net Calorific Value
1	≥ 42 MJ/kg
2	≤ 42 MJ/kg and ≥ 32 MJ/kg
3	< 32 MJ/kg

*Cooling shall be KNAN/KNAF instead of ONAN/ONAF

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BioTransol HF has been approved by UL & FM GLOBAL.

"AS TO FIRE HAZARD ONLY"
Biotransol HF
"CLASSED 4 to 5 less hazardous than paraffin oil with respect to fire hazard".
MH62641

FM APPROVED

Certificate of Compliance

This certificate is issued for the following:

Biotransol HF

Prepared for:

SAVITA POLYMERS LIMITED
66/67, Nariman Bhavan
Nariman Point
Mumbai, 400021
India

FM Approvals Class: 6933

Approval Identification: 0003061467 Approval Granted: 11/14/2017

To verify the availability of the Approved product, please refer to www.approvalguide.com or www.roofnav.com

Said Approval is subject to satisfactory field performance, continuing Surveillance Audits, and strict conformity to the constructions as shown in the Approval Guide, an online resource of FM Approvals.

Cynthia E Frank

Cynthia Frank
VP - Manager of Materials
FM Approvals
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Norwood, MA 02062

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Member of the FM Global Group

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Thermal properties of fluids

Fluid	Mass of 1000 liters	Fire point (^0C)	Specific Heat capacity (J/Kg^0C)	Temperature change (^0C)	Energy to raise to fire point (MJ)	Net Calorific Value (MJ/Kg)	Thermal Class
Mineral Oil	880 Kg	160	**1860**	70	**130.94**	46	O
Transol Synth 100	970 Kg	316	**1880**	226	**412.13**	32.6	K
Biotransol – HF	920 Kg	360	**1908**	270	**473.95**	39	K

Note:
Transformer operating temperature is considered as 90degC.

Result-

- Biotransol-HF needs awful lot of heat energy (473.95 MJ/kg), whereas mineral oil needs merely 130.94 MJ/kg to take the fluid to its fire point.

- Biotransol-HF generates less heat compared to a mineral oil during combustion (39 MJ/Kg for Biotransol-HF versus 46 MJ/Kg for mineral oil). Biotransol-HF has a self-extinguishing property while mineral oil does not possess this property.

Explosion Limits:

- The two factors discussed so far - Longer ignition time and Self-extinguishing nature of Biotransol-HF are relevant preventing fire or minimizing chances of fire within a transformer or any hot oil spilled outside a transformer.
- However the explosion inside the transformer could take place due to the formation of explosive mixture of fluid vapours with leaked in air. For an explosion to take place, one needs a very high energy arcing fault for a prolonged period of time which can potentially rise the temperature to more than 2000 – 3000 deg C. Under these conditions, concentration of fumes of the fluid in the gas phase would increase causing an explosion. This is where the explosion limits of the fluids are important

Explosive Limits	Mineral oil	Natural Ester Biotransol-HF
Test Temperature (^{0}C)	200	350 – 400
Lower Limit of flammability (% v)	0.6	> 9

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Lower Explosion Limit

- 0.6% is the LEL for mineral oil while the corresponding figure for natural ester oil is 9%.
- LEL of the ester is 15 times that of a mineral oil. Clearly, this explains why a mineral oil would explode if there is an arcing fault.
- The very high LEL makes it difficult for an ester oil to cause explosion. Thus it may be concluded safely that explosion is a highly unlikely event with Natural esters.

Advantages of using Fire safe fluid

So, what are the advantages that we are getting?

- Reduced risk to human life
- No need to evacuate surrounding area
- Space saving
- Reduced downtime costs
- Reduced insurance premium
- Reduction in Life-Cycle cost
- Especially suited for critical area of Industry.

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And now as per IEC 61936-1 –" Power installations exceeding 1 kV a.c. – Part 1: Common rules"

Table 3 – Guide values for outdoor transformer clearances

Transformer type	Liquid volume l	Clearance *G* to other transformers or non-combustible building surface m	Clearance *G* to combustible building surface m
Oil insulated transformers (O)	1 000 <…< 2 000	3	7,5
	2 000 ≤…< 20 000	5	10
	20 000 ≤…< 45 000	10	20
	≥ 45 000	15	30
Less flammable liquid insulated transformers (K) without enhanced protection	1 000 <…< 3 800	1,5	7,5
	≥ 3 800	4,5	15
Less flammable liquid insulated transformers (K) with enhanced protection	Clearance *G* to building surface or adjacent transformers		
	Horizontal m		Vertical m
	0,9		1,5

NOTE 1 Enhanced protection means

– tank rupture strength,

– tank pressure relief,

– low-current fault protection,

– high-current fault protection.

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Distance between Transformer - With **Mineral Oil**

Oil Capacity of Individual Transformer (l)	*Clear Separating Distance (m)*
>= 45000	15

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With K Class Fluids

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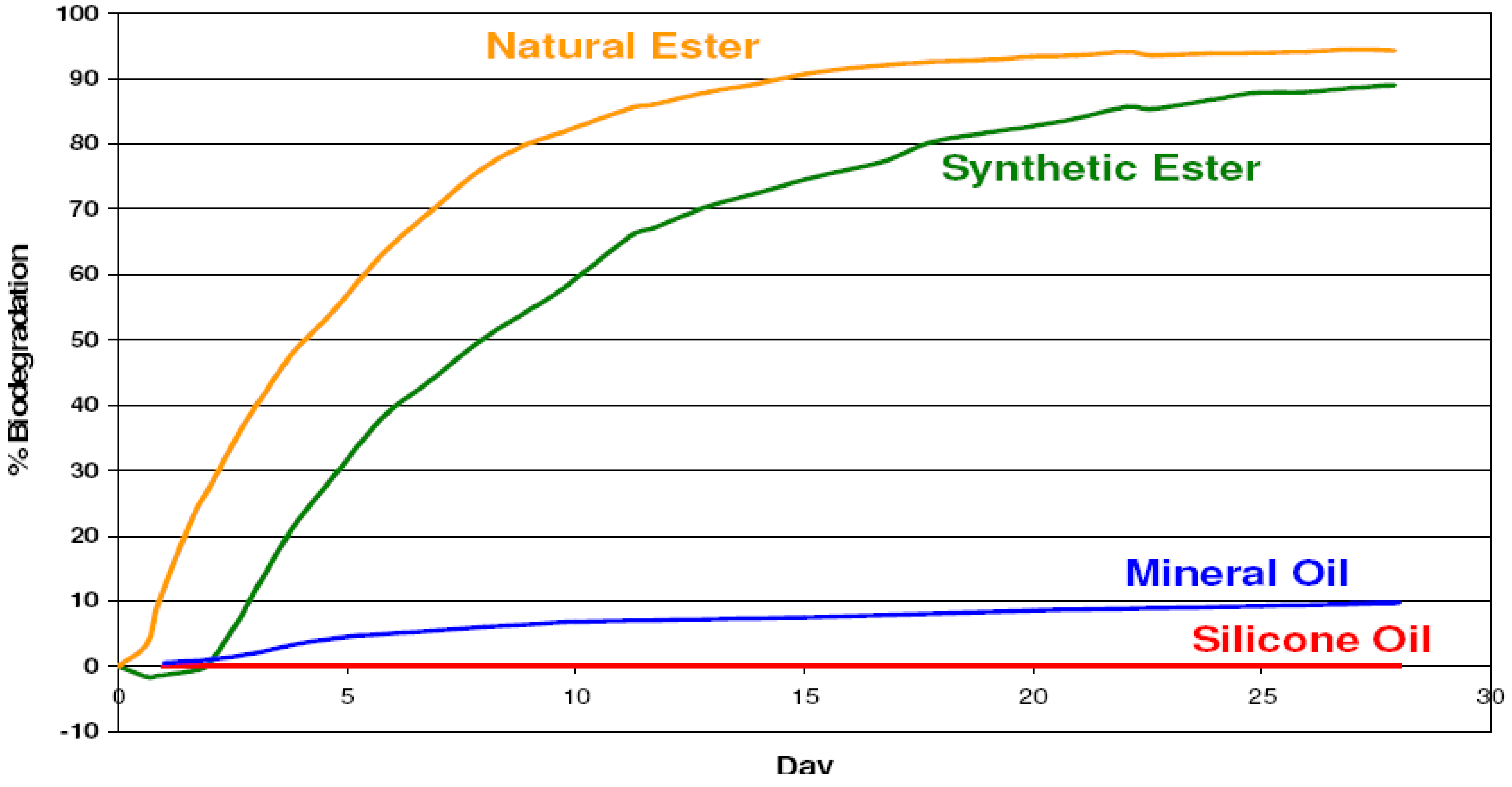
Rates of biodegradation of different insulating fluids



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Confederation of Indian Industry

CII-Green Products and Services Council

hereby certifies that

bioTRANSOL-HF

(GPSP95001)

Manufactured by Savita Polymers Limited at their plant in Mahad, Maharashtra meets the requirements of GreenPro Ecolabelling and qualifies as Green Product.

This certification is valid till ***December 2023***

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Insulation Paper life enhancement & Increase in Overloading Capacity

Transformer Life:

End of cellulose life = End of transformer life; Heat, Water and Oxygen have catalytic effect

Transformer age is determined by the tensile strength of its paper insulation. The tensile strength will decrease as the paper insulation degrades.

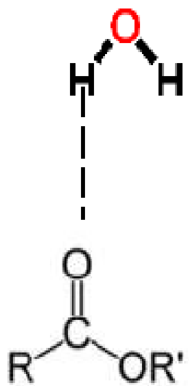
The degradation of paper insulation in transformers depends on three mechanisms:

1. **Pyrolysis (heat)**
2. **Hydrolysis (water)**
3. **Oxidation (air)**

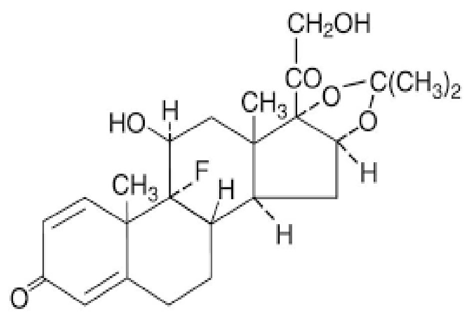
These degradation mechanisms are interrelated. Heat will cause paper to degrade. Degradation of paper will generate moisture. Moisture will contribute to the degradation of paper. The assessment of transformer life is a complex problem. The following rules of thumb exist for the three degradation mechanisms:

- Every 6°C increase in temperature above the rated hot spot temperature will result in a halving of transformer life.
- Every doubling of the moisture content in the paper insulation above 0.5% by wt will result in a halving of transformer life.
- Transformers operating in a high oxygen environment will age in the order of 2.5 times faster than transformers operating in a low oxygen environment.

Excellent Moisture tolerance



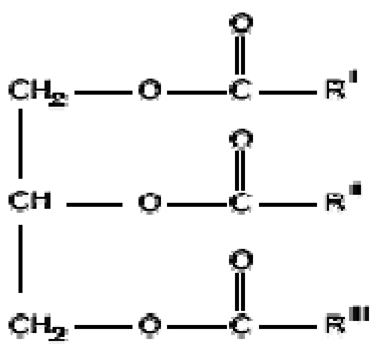
How esters attract water molecules



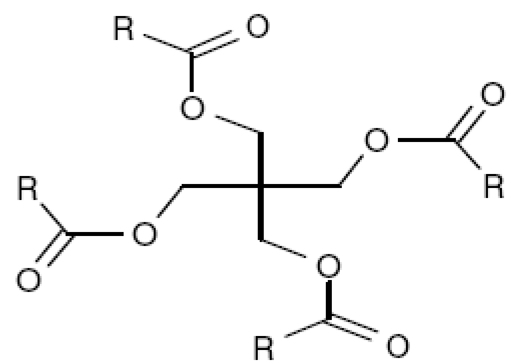
MINERAL OIL

Solubility of water in fluids

	Ester linkages	Approx water saturation at 23°C (ppm)
Mineral oil	0	55
Silicone oil	0	220
Natural ester	3	1100
Synthetic ester	4	2600



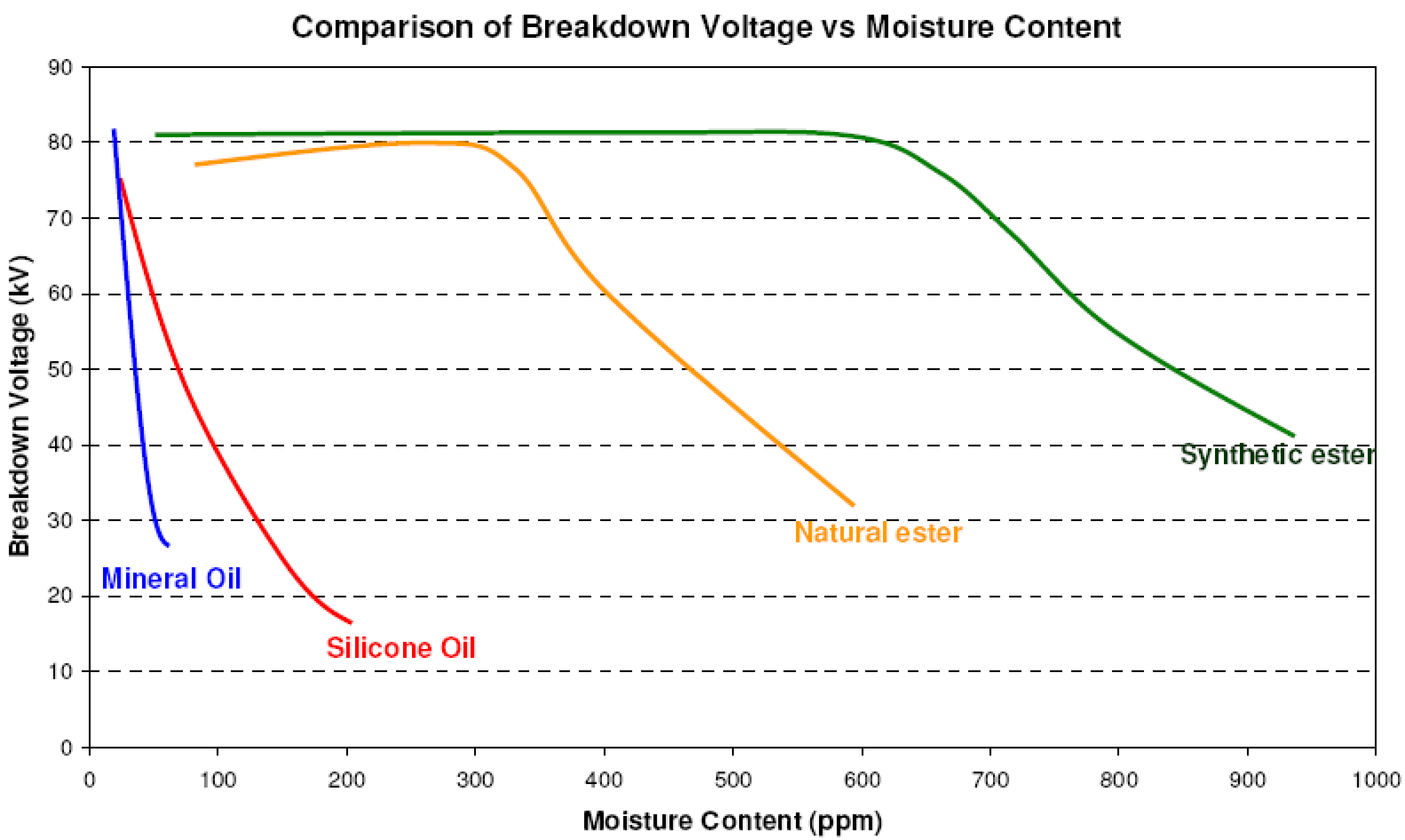
Natural Ester



Synthetic Ester

HYDROPHOBIC OR **HYDROPHILIC**

Comparison of BDV V/s. Moisture content (PPM)



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Extend life of solid insulation:

As per IEC 60076-14/IS 2026-14 the unit life of insulation is defined as

$$\text{unit life}(T) = a \times e^{\frac{15\,000}{(T+273)}}$$

where

T is the temperature in Celsius;

e is the base of the natural logarithm (2,718...);

a is a constant with the dimension hour.

Table C.2 – Comparison of ageing results

	Constant a	Temperature T °C	Thermal index	Thermal class
IEEE mineral oil/thermally upgraded paper	$9{,}80 \times 10^{-18}$	110,0	110	120
Natural ester liquid/thermally upgraded paper	$7{,}25 \times 10^{-17}$	130,6	130	140
IEEE mineral oil/kraft paper	$2{,}00 \times 10^{-18}$	95,1	95	105
Natural ester liquid/kraft paper	$1{,}06 \times 10^{-17}$	110,8	110	120

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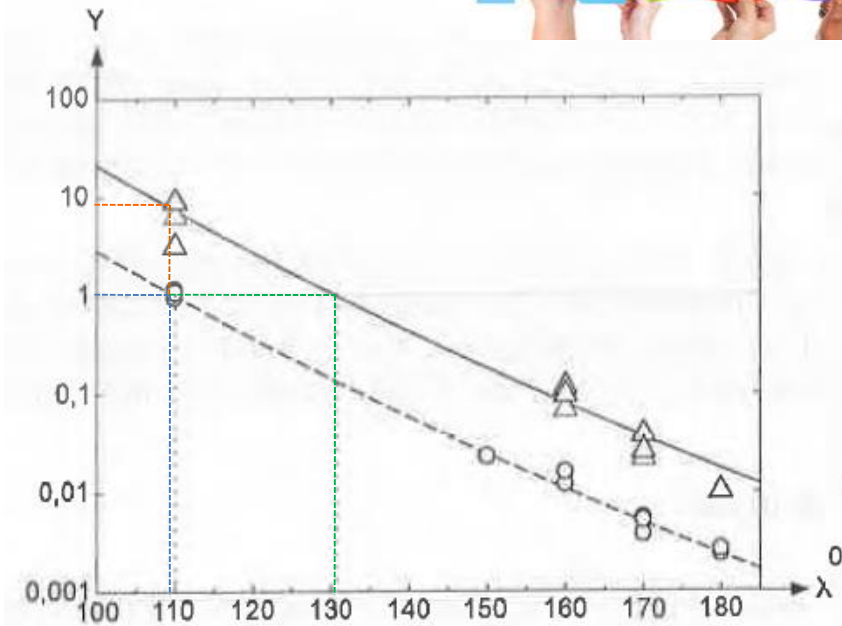

Figure C.10 – Unit life versus temperature of TUP ageing data (least squares fit)

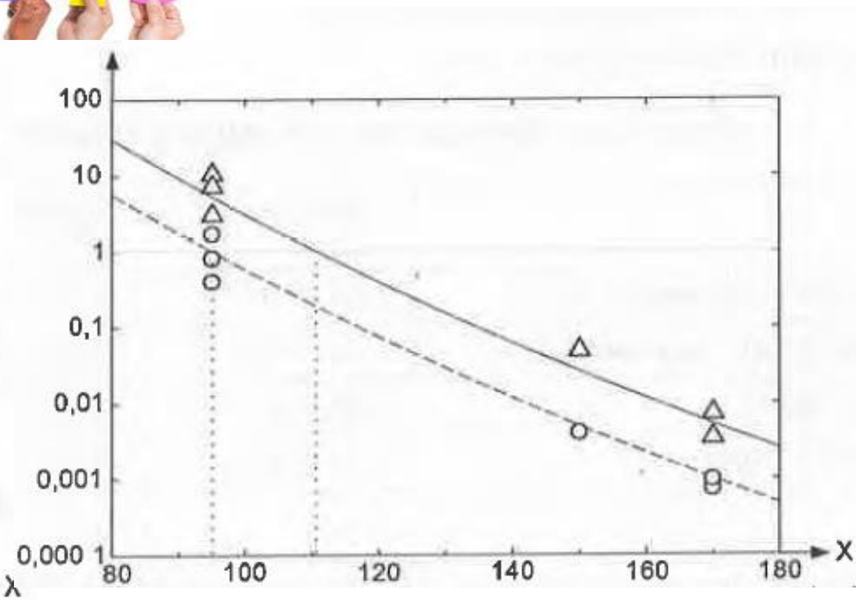

Figure C.11 – Unit life versus temperature of kraft paper ageing data (least squares fit)

•Life of the solid insulation with Natural Ester fluid will be 5 to 8 times more than MO filled Transformer.
• NE filled insulation systems can be run 20degC/15 degC warmer without degrading life i.e.. transformer can be overloaded up to 20% without sacrificing the life of Transformer.
•Existing transformers can be upgraded to potentially provide additional load capacity or additional remaining life of the transformer.

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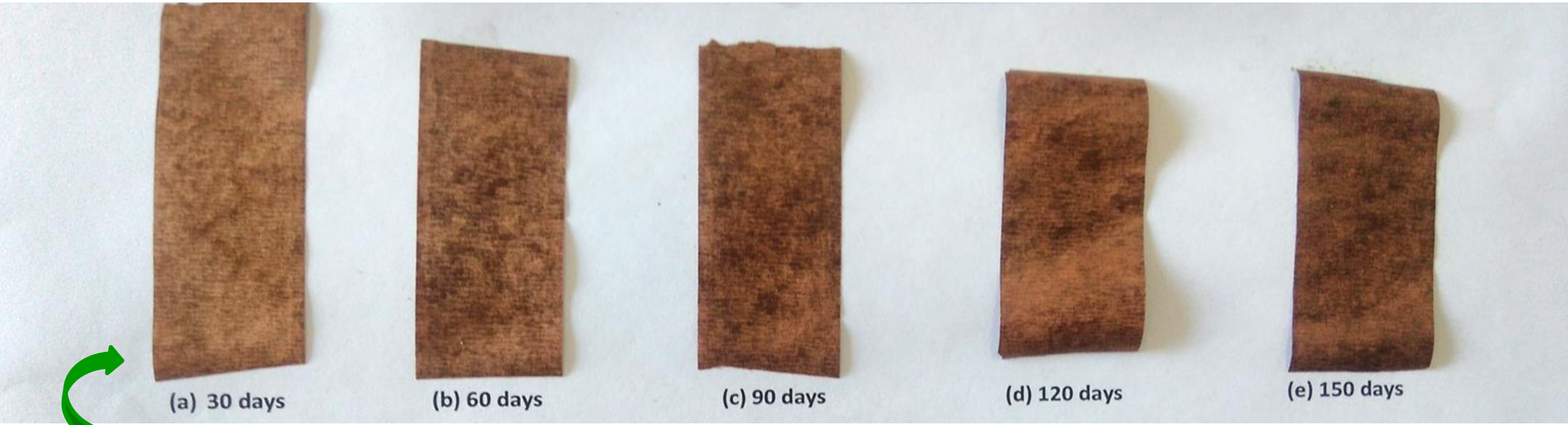
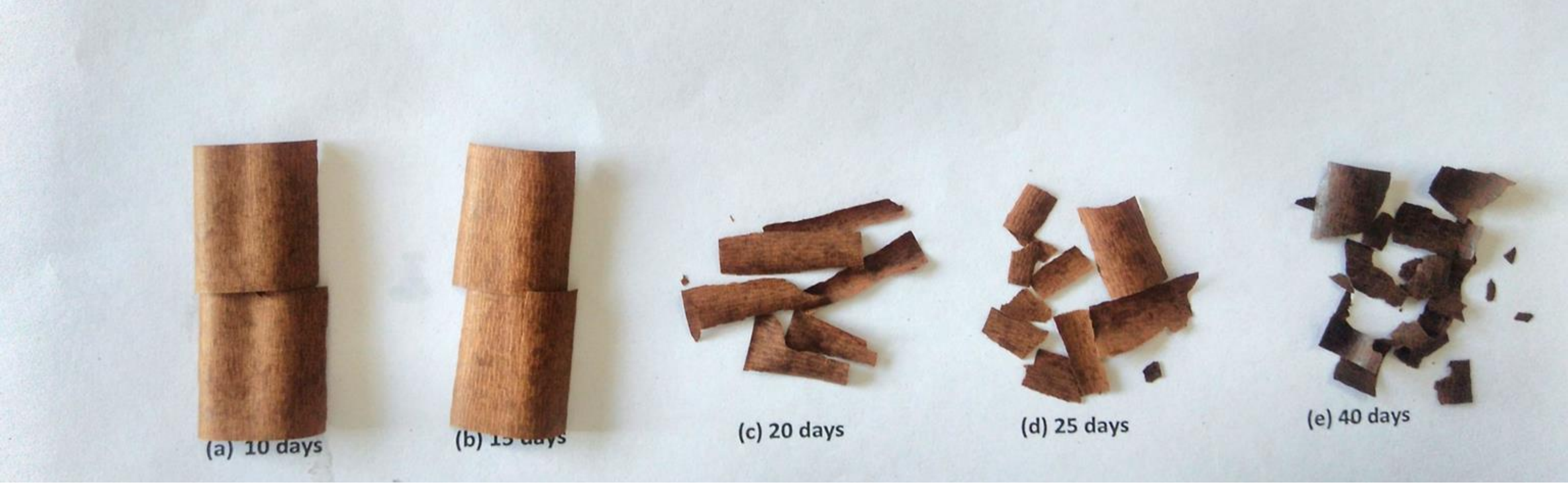
Appearance of Thermally aged TUP In Mineral Oil And Natural Ester (Biotransol HF)



Thermal aging of thermally upgraded Kraft paper in BIOTRANSOL HF at 165^{0}C



Thermal aging of thermally upgraded Kraft paper in Mineral Oil at 165^{0}C

Paper life Enhancement & Overloading

Life of the insulation paper shall enhance considerably in Natural Ester than mineral oil

Transformer with Natural Ester fluids can run 20 Deg. warmer than mineral oil without degrading the insulation paper.

Advantage is Transformer can be overloaded without damaging the transformer life with higher temperature rise limits.

The de-rating factor of Power as a function of Factor K, to allow for harmonics, is given in the below Table -

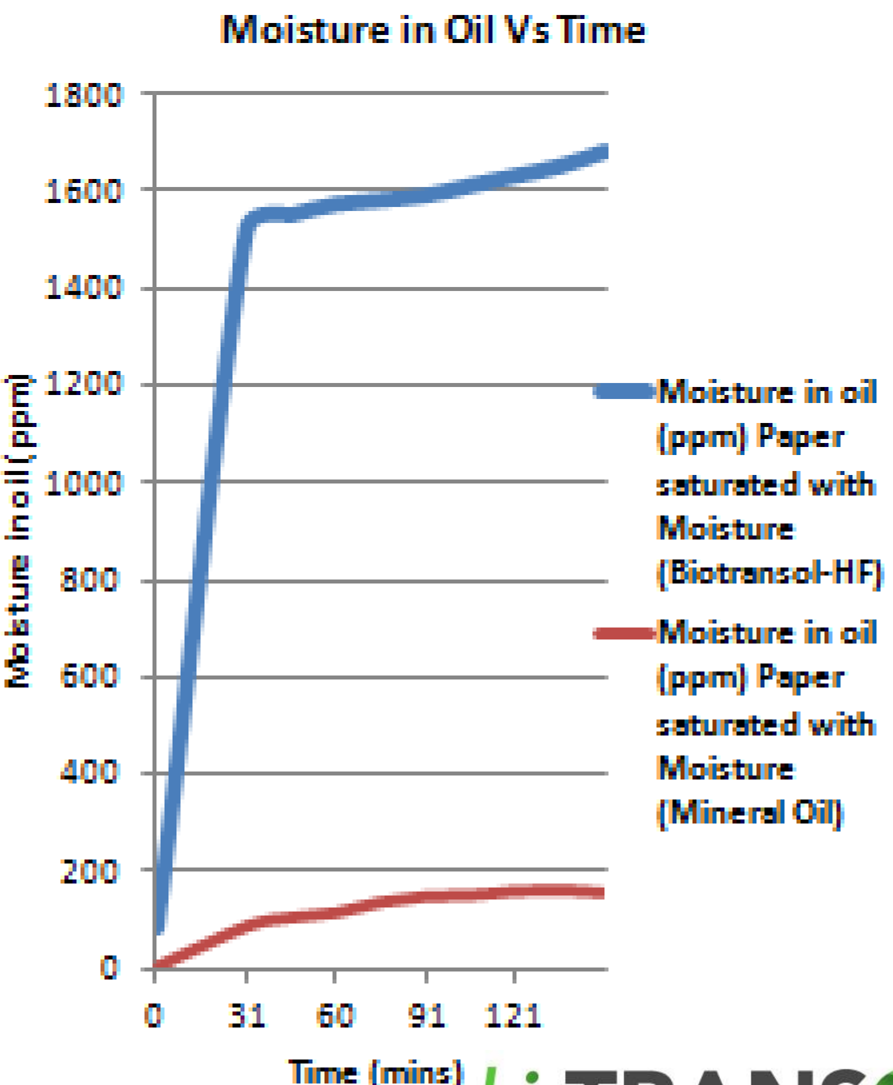
Value of factor K	*K4*	*K9*	*K13*	*K20*	*K30*	*K40*	*K50*
Derating factor	0,886	0,761	0,692	0,606	0,524	0,469	0,428

The most common effects of nonlinear loads on distribution transformers as per IEEE C57.110 are:

•Saturation of transformer's core by changing its operating point towards the knee of the nonlinear B-H curve,
•Increase in core (hysteresis and eddy current) power losses,
•Increase in fundamental and harmonic copper losses,
•Increase in the temperature of windings, cleats, leads, insulation and oil, which can cause overheating,
•Bushings, tap changers and cable-end connections will also be exposed to higher stresses, which can result in transformer's failure,
•Transformer's efficiency reduction and power factor decrease,
•Transformer's de-rating,
•Reduction of transformer's life-expectancy etc.

DRYING OF PAPER BY OIL

Time (min)	Moisture in oil (ppm)	
	Paper saturated with Moisture (Biotransol-HF)	Paper saturated with Moisture (Mineral Oil)
0	87	3
30	1520	87
45	1550	105
60	1572	115
75	1580	137
90	1590	149
105	1610	151
120	1630	160
135	1650	162
150	1680	157



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WHY?

- These answer are clearly mentioned is IEC 60076-14/IS 2026-14.

- The Water reacts with the triglycerides of Natural Ester and form **Long Chain Fatty Acid** through a process called <u>Trans-esterification.</u>

- Three Water Molecule are needed to add –H & -OH group to break the Ester Bond i.e. reduction of Water PPM level in Insulating Oil.

- The Long Chain Fatty Acid then bond to the cellulose structure and appear to form a barrier to water ingress and with that a decline in the rate of deterioration of the cellulose insulation.

- The <u>total Acid number</u> in a aged Natural Ester filled transformer will be <u>higher than Mineral oil filled Transformer</u>. However the acid formed i.e. *Long Chain Fatty Acid seem to be non-corrosive in nature* compared to the *short-chain fatty acid, which is corrosive in nature*, in case of Mineral Oil filled Transformer.

NO SLUDGE

General Design considerations while using Ester fluids

- Natural Ester fluid has to be used in sealed transformers eg. Aircell in the conservator.
- Ester fluids has higher viscosity than mineral oil which is inversely proportional with Operating Temperatures.
- Ester has higher specific heat & better thermal conductivity than mineral oil which compensates the higher viscosity heat dissipation limitations.

Offerings of Natural Ester fluid

- Environment preservation
- Enhanced Fire Safety
- Asset life extension
- Less Maintenance
- Overloading Capacity
- Supporting Make in India
- Supporting to Agricultural and allied Industry

National & International Standards:

IEC 60076-14 & IS 2026-14 – Liquid immersed power Transformers using High Temperature insulation material.

IEC 60695-1-40 – Guidance for assessing the fire hazard of electro technical products – Insulating liquids -'4 Classification of insulating liquids-Table 1

IEC 61936-1 - Power installations exceeding 1 kV a.c. –Part 1: Common rules - Table 3 & 4 – Guide values for outdoor & indoor transformer clearances with Less flammable liquid insulated transformers (K)

IS 13503:2013 - Classification of Insulating Liquids

IEC 62770 – Fluids for electromechanical applications – Unused natural esters for transformers and similar electrical equipment.

IS 16659 - Fluids for electromechanical applications – Unused natural esters for transformers and similar electrical equipment.

IEC 61203:1992 - Synthetic organic esters for electrical purposes - Guide for maintenance of transformer esters in equipment

IS 16081/IEC 61099 – Specifications for unused Synthetic Organic Esters for Electrical Purposes

IEEE C57.147 - Guide for Acceptance and Maintenance of Natural Ester Fluids in Transformers- BIS Standard **ETD 3 (11123)**

IEC 62975 – Natural Ester: Guideline for maintenance and use in Electrical Equipment

IS 1180 Part 3 – Outdoor type , insulated liquid immersed distribution transformer Up to and Including 2500 KVA, 33 kV–specification

IEEE-C57-155/IS 16785 - IEEE Guide for Interpretation of Gases Generated in Natural Ester and Synthetic Ester-Immersed Transformers

IEEE-C57-201- IEEE Standard for the design, testing & application Of liquid immersed Distribution, Power & regulating Transformers Using High Temperature Insulation Systems & Operating at elevated Temperatures .

C57.100-2011 - IEEE Standard Test Procedure for thermal evaluation of Insulation System for Liquid-Immersed Distribution & Power Systems

D6871-03 - Standard Specification for Natural Ester Fluids Used In Electrical Apparatus

CEA Standard Specification and technical parameters for Transformers and Reactors (66 KV and above voltage class)

Savita Credentials

We are proud to inform that Savita has supplied bioTRANSOL Natural Ester fluid to M/s. Atlanta Electricals for 100 MVA, 220/66 KV Power Transformer supplied to GETCO (Gujarat Energy Transmission Corporation Limited)

This is the first highest rating, highest voltage class Natural Ester filled Transformer manufactured in India for an Indian Utility till date.

Also bioTRANSOL has supplied to M/S Schneider Electric for 55 MVA, 34.5/11 KV Generator Transformer .

This is the first highest rating GT with NE Oil and in operation for more than a year.

Testing & Certification of Natural Ester fluid

- Our Ester fluids are type tested at CPRI, ERDA & Laborelec-Belgium as per latest IS/IEC Standards
- Biodegradability test as per OECD method
- FM Global Approved
- UL Approved
- GRIHA (Green Ratings for Integrated Habitat Assessment by MNRE & TERI) approved
- GREENPRO by IGBC approved

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bioTRANSOL Approval by Transformer OEMs

- GE (France)
- ABB (Sweden)
- Schneider Electric (France)
- CG Industrial Solutions
- Transformer & Rectifiers (I) Limited
- TBEA
- Atlanta Electricals Limited
- Voltamp
- Raychem RPG

and many more….

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biOTRANSOL Approval by Accessories Manufacturers

- MR Germany(OLTC),
- CAPT, Italy
- Easun MR
- HMG (OLTC),
- CTR (OLTC),
- OLG (OLTC),
- **Material compatibility tests** conducted as per ASTM on

  Core,

  Copper,

  Pressboards,

  Fiberglass,

  Nitrile Rubber,

  Permawood Strip, Paint…….

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Special Testing with OEMs

- Tests conducted at CG R&D laboratory
  HV test for different insulation classes,
  LV test for different insulation classes,
  Thermal & electrical properties of different oil gaps.

- Oil tested at CTR R&D NABL lab, for 132,220 & 400 KV class Tap Changers

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OUR FIELD EXPERIENCE

- We have the largest fluid condition monitoring experience in India. (More than 1000 samples of different periods)

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GETCO

Transformers & Rectifiers (India) make 20 MVA, 66/11 KV Power Transformer filled with bioTRANSOL Natural Ester fluid installed at Gujarat Energy Transmission Corporation Limited (State utility in India) substation.

This is India's first largest project with Natural ester fluid filled Power Transformers. Savita has supplied bioTRANSOL Natural Ester fluid for total 110 Nos. 15 & 20 MVA Transformers to M/s. Atlanta Electrics Limited, M/s. Transformers & Rectifiers (I) Limited & M/s. Voltamp Limited.

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FLUID MONITORING FIELD EXPERIENCE AT REGULAR INTERVALS FOR GETCO

a. Physical Properties like kinematic viscosity

b. Electrical Properties like BDV, Dielectric Dissipation factor.

c. Chemical properties like Water content, Soluble acidity

d. Health, safety & environment like flash point, fire point & DGA

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Natural Ester Fluid Parameters for in service Transformer

Ref: IEC 62975

Test	Value under 'Good' category
	≤ 69 kV
Dissipation factor (power factor), ASTM D924, at 25 °C, %	< 0.15
Breakdown Voltage in kV	>40
Water content, mg/kg	< 200
Viscosity increase from value at time of initial energization, ASTM D445, at 40 °C, %	< 10
Acid number, mg KOH/gm	< 0.3
Flash point, °C	> 275
Color	Clear and without visible contamination

Transformer -1

Name of Customer – T & R, GETCO

Properties	Bf Energ	Aft Ener g	1 M	2 M	3 M	4 M	5 M	6 M	10 M	12 M	22 M	24 M	30 M	36 M
Soluble Acidity, mg of KOH.goil^{-1}	0.04	0.04	0.041	0.042	0.04	0.042	0.044	0.046	0.043	0.047	0.045	0.042	0.044	0.045
Kinematic Viscosity at 40degC mm^2.S^{-1}	34.5	34.5	34.6	34.5	34.4	34.5	34.4	34.6	34.3	34.5	34.6	34.4	34.5	34.5
Flash Point (deg C)	268	268	267	267	264	265	266	263	265	266	268	267	268	267
Fire Point (deg C)	358	360	359	360	359	359	359	358	358	357	358	357	357	357
BDV (kV)	78	76	82	87	90	85	83	84	86	83	84	82	83	84
Dissipation factor @ 90 deg C	0.041	0.039	0.036	0.034	0.038	0.034	0.037	0.039	0.041	0.044	0.041	0.043	0.045	0.043

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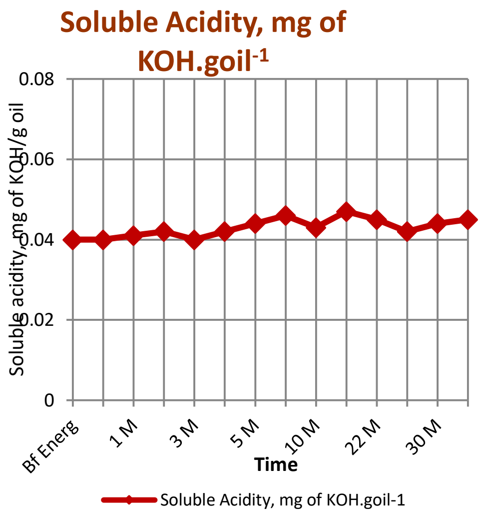

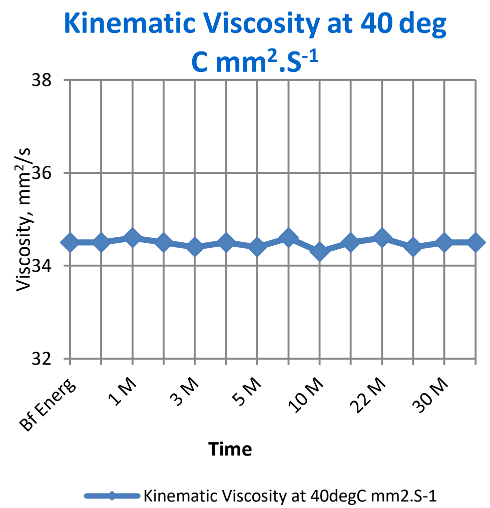

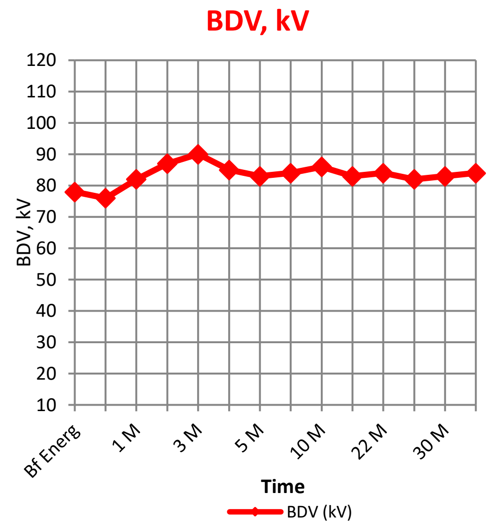

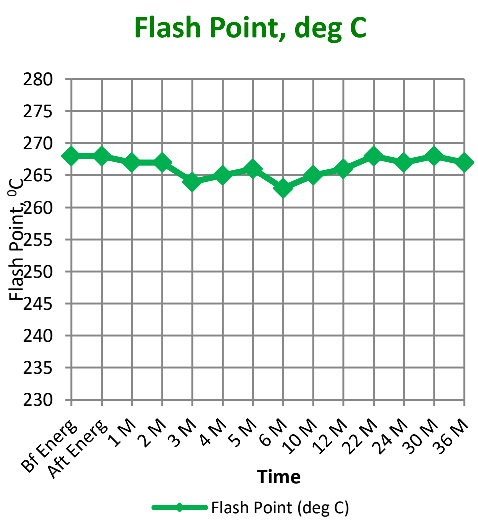

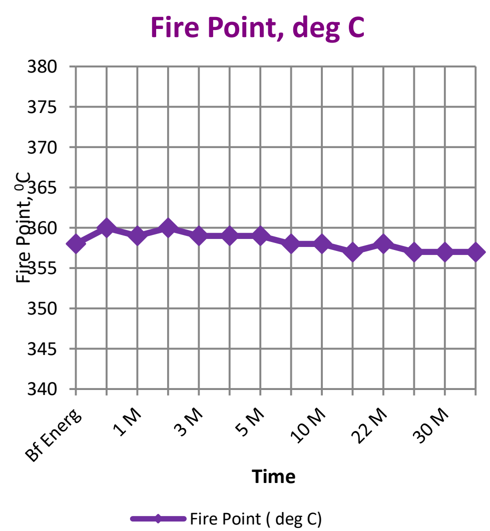

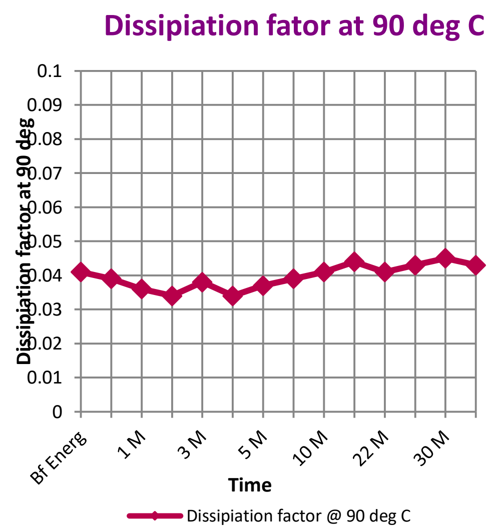

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DGA Results

SAMPLE DURATION	$CO_{2[PPM]}$	$C_2H_{4[PPM]}$	$C_2H_{2[PPM]}$	$C_2H_{6[PPM]}$	$H_{2[PPM]}$	$CH_{4[PPM]}$	$CO_{[PPM]}$
		Ethylene	Acetylene	Ethane		Methane	
Before Energizing	66.9	0.0	0.0	11.1	0.0	0.0	7.5
24 hrs. after energizing	101.6	0.0	0.0	10.0	0.0	0.0	14.7
1 M	130.9	ND	0.0	10.9	0.0	ND	31.8
2 M	264.4	ND	0.0	11.3	0.0	ND	48.7
3 M	48.2	0.0	0.0	13.0	0.0	0.0	12.8
4 M	29.3	ND	0.0	12.4	0.0	ND	14.3
5 M	335.2	ND	0.0	15.0	0.0	1.1	26.8
6 M	510.4	1.4	0.0	19.2	0.0	2.1	76.5
10 M	446.7	1.3	0.0	21.2	0.0	1.3	53.1
12 M	315.1	1.2	0.0	23.5	0.0	1.3	25.7
22M	270.98	1.23	0.00	21.32	0.00	1.53	34.40
24M	384.70	1.29	0.00	20.89	0.00	1.55	28.33
30M	336.06	2.90	0.00	27.74	0.00	0.00	36.78
36M	203.10	1.83	0.00	33.40	0.00	0.00	22.34

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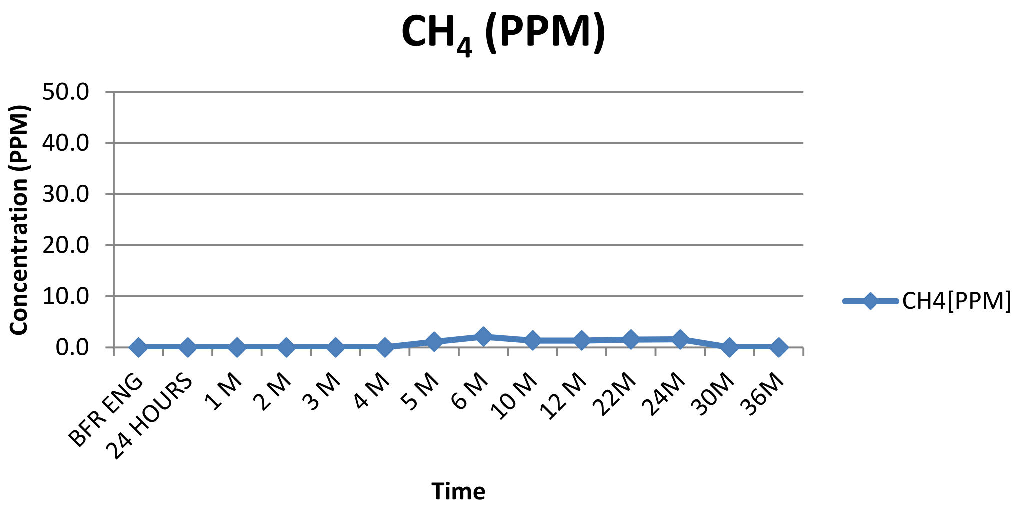
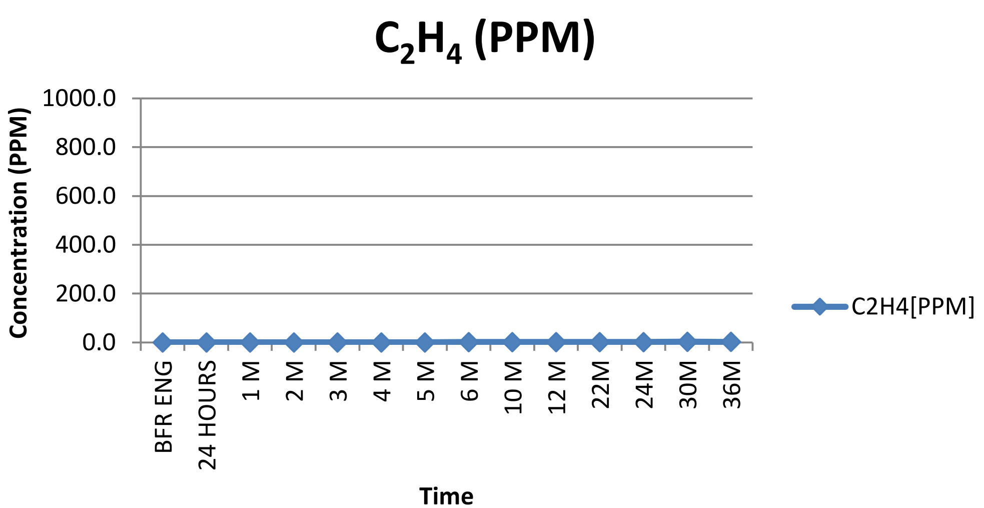
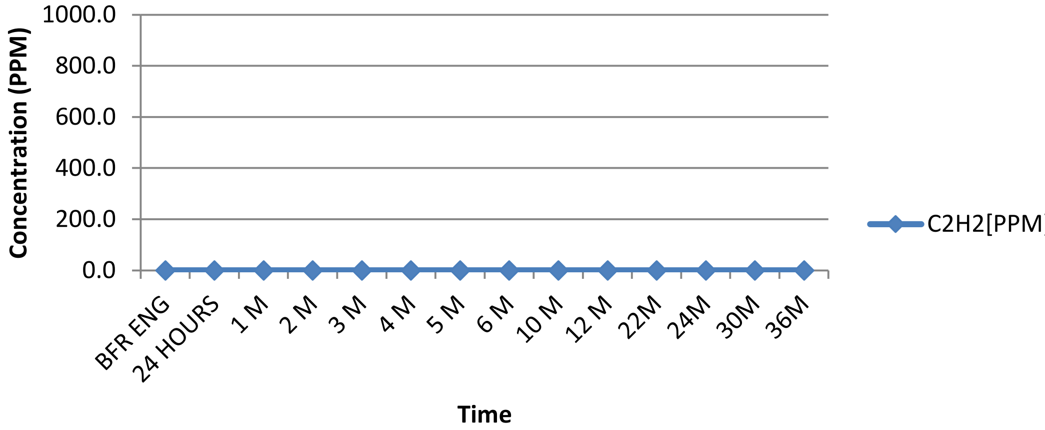


CH4 (PPM)
Concentration (PPM)
50.0
40.0
30.0
20.0
10.0
0.0
BFR ENG
24 HOURS
1 M
2 M
3 M
4 M
5 M
6 M
10 M
12 M
22M
24M
30M
36M
Time
CH4[PPM]

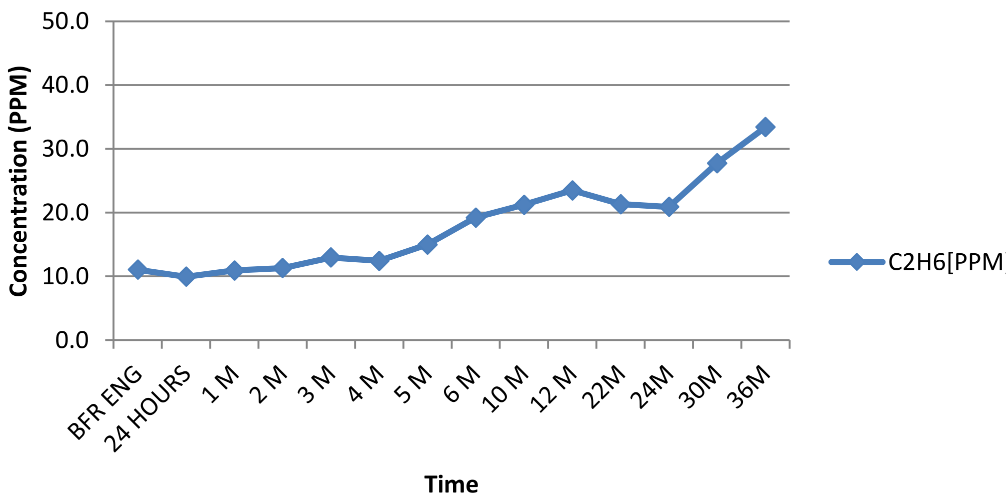


C2H4 (PPM)
Concentration (PPM)
1000.0
800.0
600.0
400.0
200.0
0.0
BFR ENG
24 HOURS
1 M
2 M
3 M
4 M
5 M
6 M
10 M
12 M
22M
24M
30M
36M
Time
C2H4[PPM]

C_2H_2 (PPM)



C_2H_6 (PPM)



Transformer - 2

Name of Customer – T & R, GETCO

Properties	Bf Energ	24 hrs	1 week	1 M	2 M	3 M	4 M	5 M	6 M	10 M	12 M	20 M	30 M
Soluble Acidity, mg of KOH.goil^{-1}	0.044	0.042	0.04	0.042	0.044	0.042	0.048	0.041	0.046	0.043	0.041	0.044	0.042
Kinematic Viscosity at 40degC $mm^2.S^{-1}$	34.6	34.7	34.6	34.5	34.3	34.4	34.2	34.3	34.5	34.7	34.5	34.6	34.5
Flash Point (deg C)	266	265	265	266	265	266	266	266	264	266	268	268	267
Fire Point (deg C)	359	359	360	360	360	360	359	359	357	358	360	358	358
BDV (kV)	80	82	79	77	81	83	78	81	82	85	86	83	82
Dissipation factor @ 90 deg C	0.04	0.038	0.041	0.043	0.041	0.043	0.048	0.045	0.046	0.042	0.046	0.044	0.046

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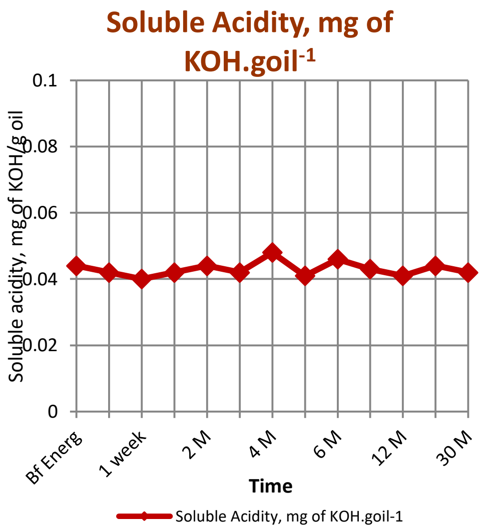

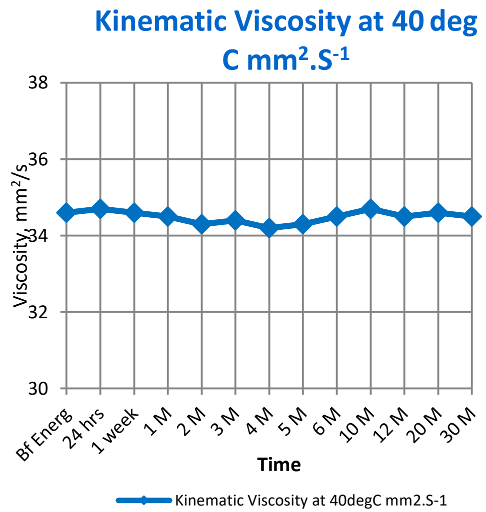

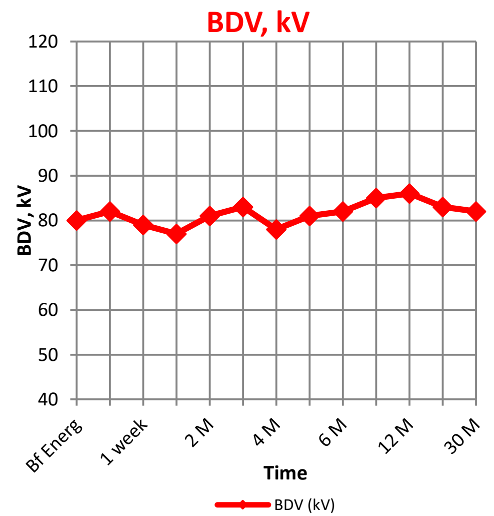

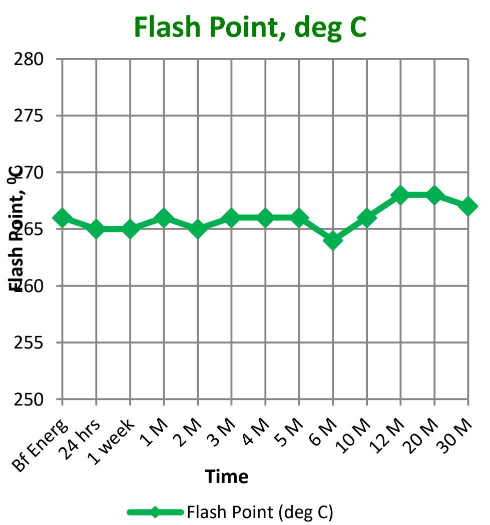

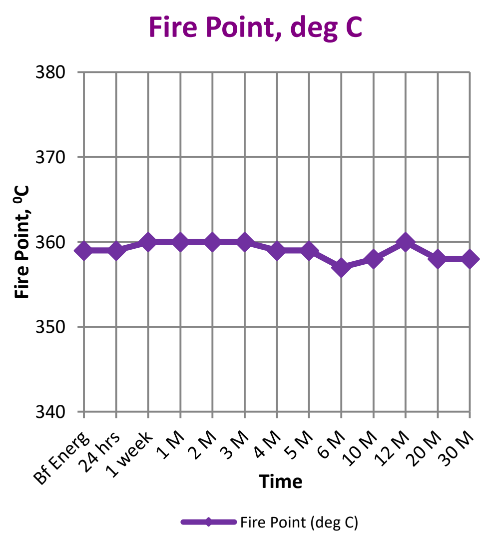

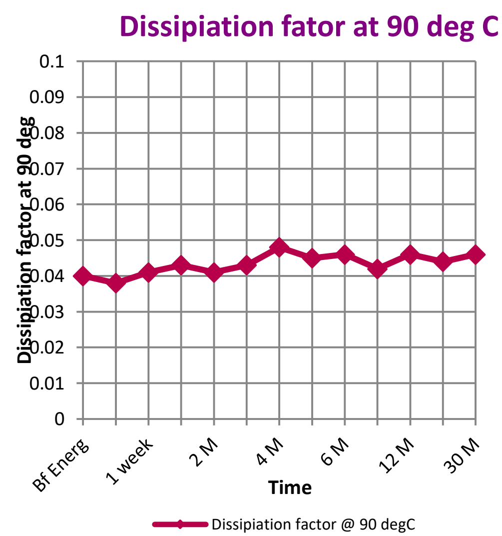

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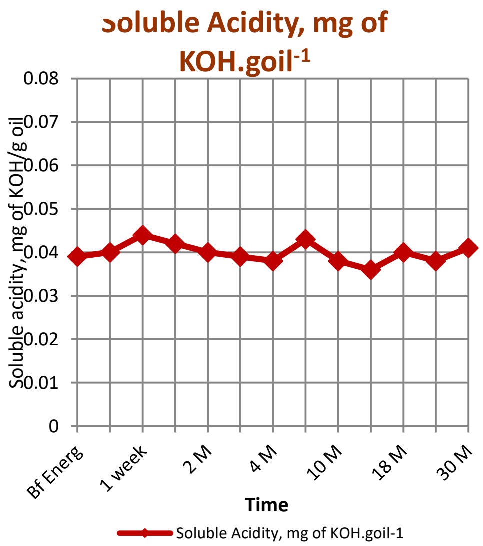
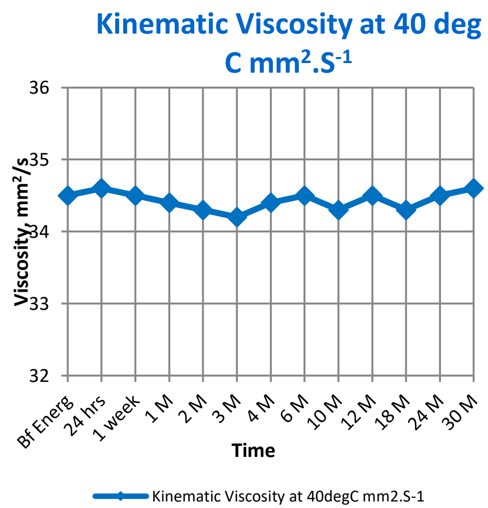
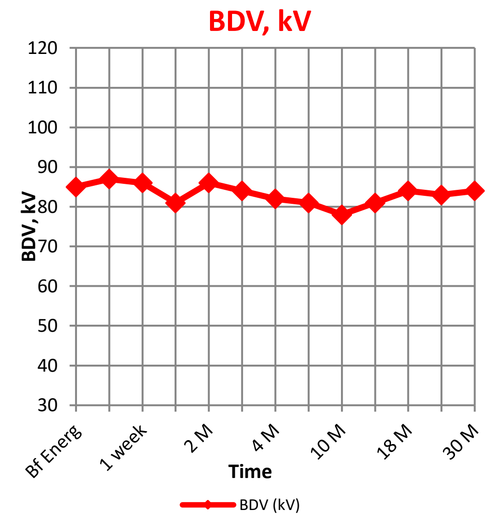
Transformer - 3

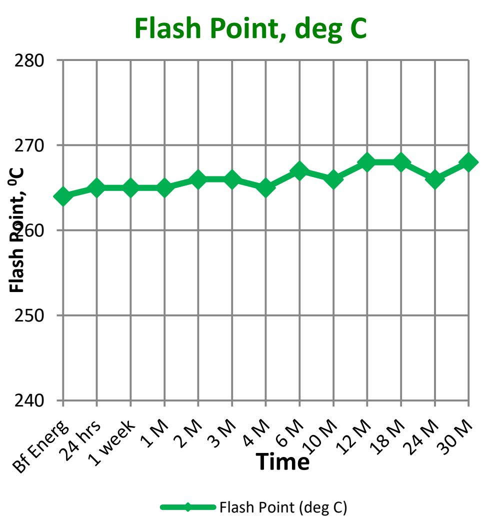
Name of Customer – T & R, GETCO

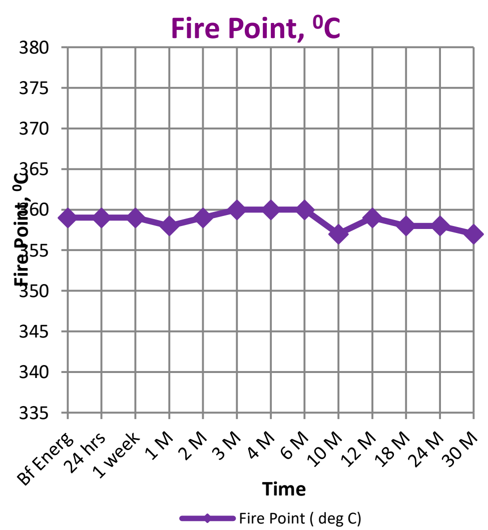
Properties	Bf Energ	24 hrs	1 week	1 M	2 M	3 M	4 M	6 M	10 M	12 M	18 M	24 M	30 M
Soluble Acidity, mg of KOH.goil^{-1}	0.039	0.04	0.044	0.042	0.04	0.039	0.038	0.043	0.038	0.036	0.04	0.038	0.041
Kinematic Viscosity at 40degC mm^2.S^{-1}	34.5	34.6	34.5	34.4	34.3	34.2	34.4	34.5	34.3	34.5	34.3	34.5	34.6
Flash Point (deg C)	264	265	265	265	266	266	265	267	266	268	268	266	268
Fire Point (deg C)	359	359	359	358	359	360	360	360	357	359	358	358	357
BDV (kV)	85	87	86	81	86	84	82	81	78	81	84	83	84
Dissipation factor @ 90 deg C	0.029	0.031	0.027	0.032	0.034	0.038	0.036	0.041	0.044	0.041	0.045	0.042	0.044

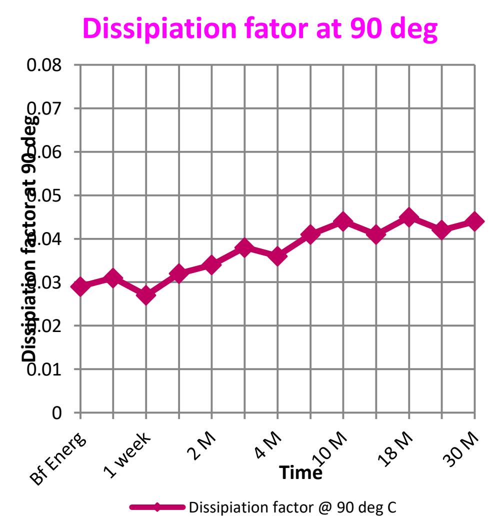

Soluble Acidity, mg of KOH.goil-1
Soluble acidity, mg of KOH/g oil
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
Bf Energ
1 week
2 M
4 M
10 M
18 M
30 M
Time
Soluble Acidity, mg of KOH.goil-1


Kinematic Viscosity at 40 deg C mm2.S-1
Viscosity, mm2/s
36
35
34
33
32
Bf Energ
24 hrs
1 week
1 M
2 M
3 M
4 M
6 M
10 M
12 M
18 M
24 M
30 M
Time
Kinematic Viscosity at 40degC mm2.S-1


BDV, kV
BDV, kV
120
110
100
90
80
70
60
50
40
30
Bf Energ
1 week
2 M
4 M
10 M
18 M
30 M
Time
BDV (kV)


Flash Point, deg C
Flash Point, 0C
280
270
260
250
240
Bf Energ
24 hrs
1 week
1 M
2 M
3 M
4 M
6 M
10 M
12 M
18 M
24 M
30 M
Time
Flash Point (deg C)


Fire Point, 0C
Fire Point, 0C
380
375
370
365
360
355
350
345
340
335
Bf Energ
24 hrs
1 week
1 M
2 M
3 M
4 M
6 M
10 M
12 M
18 M
24 M
30 M
Time
Fire Point (deg C)


Dissipiation fator at 90 deg
Dissipiation factor at 90 deg
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
Bf Energ
1 week
2 M
4 M
10 M
18 M
30 M
Time
Dissipiation factor @ 90 deg C

THANK YOU!